Additional Exercises

Moments of Normal Dist^n

$X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}} \; ; \quad -\infty < x < \infty, \; -\infty < \mu < \infty, \; \sigma^2 > 0$$

Odd order central moments

$$\mu_{2n+1} = \int_{-\infty}^{\infty} (x-\mu)^{2n+1} f(x)\,dx$$

$$= \int_{-\infty}^{\infty} (x-\mu)^{2n+1} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}\,dx$$

$$z = \frac{x-\mu}{\sigma} \qquad \sigma\,dz = dx$$

$$= \int_{-\infty}^{\infty} (\sigma z)^{2n+1} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,dz = \frac{\sigma^{2n+1}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n+1} e^{-\frac{z^2}{2}}\,dz = 0$$

Since the integrand $z^{2n+1} e^{-\frac{z^2}{2}}$ is an odd function of z.

Even order central moments

$$\mu_{2n} = \int_{-\infty}^{\infty} (x-\mu)^{2n} f(x)\,dx = \int_{-\infty}^{\infty} (\sigma z)^{2n} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,dz = \frac{\sigma^{2n}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n} e^{-\frac{z^2}{2}}\,dz$$

$$= \frac{2\sigma^{2n}}{\sqrt{2\pi}} \int_0^{\infty} z^{2n} e^{-\frac{z^2}{2}}\,dz$$

(since integrand is an even function of z)

Let $\frac{z^2}{2} = t \Rightarrow 2z\,dz = 2dt$, $dz = \frac{dt}{z} = \frac{dt}{\sqrt{2t}}$

$$= \frac{2\sigma^{2n}}{\sqrt{2\pi}} \int_0^{\infty} (2t)^n e^{-t} \frac{dt}{\sqrt{2t}}$$

$$= \frac{2^n \sigma^{2n}}{\sqrt{\pi}} \int_0^{\infty} e^{-t}\, t^{(n+\frac{1}{2})-1}\,dt$$

$$\Rightarrow \boxed{\mu_{2n} = \frac{2^n \sigma^{2n}}{\sqrt{\pi}} \Gamma\left(n+\tfrac{1}{2}\right)}$$

Changing n to $n-1$, we get.

$$\mu_{2n-2} = \frac{2^{n-1}\sigma^{2n-2}}{\sqrt{\pi}} \Gamma\left(n-\tfrac{1}{2}\right)$$

$$\frac{\mu_{2n}}{\mu_{2n-2}} = 2\sigma^2 \frac{\Gamma\left(n+\frac{1}{2}\right)}{\Gamma\left(n-\frac{1}{2}\right)} = 2\sigma^2\left(n-\tfrac{1}{2}\right) = 2\sigma^2 \frac{(2n-1)}{2} = \sigma^2(2n-1)$$

$$\Gamma\left(n+\tfrac{1}{2}\right) = \Gamma\left(n+\tfrac{1}{2}-1+1\right) = \Gamma\left(\left(n-\tfrac{1}{2}\right)+1\right) = \left(n-\tfrac{1}{2}\right)\Gamma\left(n-\tfrac{1}{2}\right)$$

$$\Rightarrow \boxed{\mu_{2n} = \sigma^2(2n-1)\,\mu_{2n-2}} \text{ — recurrence relation}$$

We know $\mu_{2n} = \sigma^2(2n-1)\mu_{2n-2}$

$$= \sigma^2(2n-1)\left[(2n-3)\sigma^2\right]\mu_{2n-4}$$

$$= \sigma^2(2n-1)\left[(2n-3)\sigma^2\right]\left[(2n-5)\sigma^2\right]\mu_{2n-6}$$

$$\vdots$$

$$\left[(2n-1)\sigma^2\right]\left[(2n-3)\sigma^2\right]\left[(2n-5)\sigma^2\right]\cdots(3\sigma^2)(1.\sigma^2)$$

$$= 1.3.5\cdots(2n-1)\sigma^{2n}$$

Ex. $X \sim N(\mu, \sigma^2)$

$Y = \frac{1}{2}\left[\frac{X-\mu}{\sigma}\right]^2$, Find mean and variance of Y.

$Y = \frac{1}{2}Z^2$ $\quad Z^2 \sim \chi^2_1$ $\quad$ Mean $= 1$ variance 2.

$E(Y) = \frac{1}{2}E(Z^2) = \frac{1}{2}\cdot 1 = \frac{1}{2}$

$V(Y) = \frac{1}{4}V(Z^2) = \frac{1}{4}\cdot 2 = \frac{1}{2}$

Ex. $X \sim N(0,1)$ $\quad X^2 \sim \chi^2_1$, $V(X^2) = 2$

Find $V(X^2)$

Ex. $X \sim N(0,1)$ $\quad Y \sim N(0,1)$

$Z = X^2 + Y^2 \sim \chi^2_2$

$E(Z) = 2$, $V(Z) = 4$

1)

Ex. $X \sim \beta_1(\mu,\nu)$ and $Y \sim \Gamma(\lambda, \mu+\nu)$ be indep. r.vs ($\mu,\nu,\lambda>0$). Find the p.d.f. of XY and identify the distribution.

Sol^n The joint p.d.f. of X and Y is

$$f(x,y) = \frac{1}{B(\mu,\nu)} x^{\mu-1}(1-x)^{\nu-1} \cdot \frac{\lambda^{\mu+\nu}}{\Gamma(\mu+\nu)} e^{-\lambda y} y^{\mu+\nu-1} \; ; \; 0<x<1, \; 0<y<\infty$$

1.σ²)

Consider the transformation

$$x \cdot y = u, \quad x = z \qquad \text{i.e} \quad x = z \text{ and } y = \frac{u}{z}$$

$$J = \frac{\partial(x,y)}{\partial(u,z)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial z} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ \frac{1}{z} & -\frac{u}{z^2} \end{vmatrix} = -\frac{1}{z}$$

$$|J| = \frac{1}{z}$$

$$g(u,z) = \frac{\lambda^{\mu+\nu}}{B(\mu,\nu)\Gamma(\mu+\nu)} z^{\mu-1}(1-z)^{\nu-1} e^{-\lambda\frac{u}{z}} \left(\frac{u}{z}\right)^{\mu+\nu-1} \cdot \frac{1}{z}$$

ce 2.

$$0<z<1, \quad 0<u<\infty$$

$$g_1(u) = \frac{\lambda^{\mu+\nu}}{B(\mu,\nu)\Gamma(\mu+\nu)} u^{\mu+\nu-1} \int_0^1 \frac{(1-z)^{\nu-1}}{z^{\nu+1}} e^{-\lambda\frac{u}{z}} dz$$

$$= \frac{\lambda^{\mu+\nu} u^{\mu+\nu-1}}{\cancel{B(\mu,\nu)}\,\Gamma\mu\,\Gamma\nu} \int_0^1 \frac{1}{z^2}\left(\frac{1}{z}-1\right)^{\nu-1} e^{-\lambda\frac{u}{z}} dz$$

$$= \frac{\lambda^{\mu+\nu} u^{\mu+\nu-1}}{\Gamma\mu\,\Gamma\nu} \int_0^\infty t^{\nu-1} e^{-\lambda u(1+t)} dt \qquad \left(\frac{1}{z}-1 = t\right.$$

$$\frac{dt}{dz} = -\frac{1}{z^2}$$

$$dz = -z^2 dt$$

$$= \frac{\lambda^{\mu+\nu} u^{\mu+\nu-1} e^{-\lambda u}}{\Gamma\mu\,\Gamma\nu} \int_0^\infty e^{-\lambda u t} t^{\nu-1} dt$$

$$= \frac{\lambda^{\mu+\nu} u^{\mu+\nu-1}}{\Gamma\mu\,\cancel{\Gamma\nu}} e^{-\lambda u} \frac{\cancel{\Gamma\nu}}{(\lambda u)^\nu}$$

$$= \frac{\lambda^\mu}{\Gamma\mu} e^{-\lambda u} u^{\mu-1}, \quad 0<u<\infty$$

$$= \Gamma(\lambda,\mu) \qquad X \cdot Y \sim \Gamma(\lambda,\mu)$$

Ex. In an airport, five radars are in operation and each radar has a 0.9 prob. of detecting an arriving airplane. The radars operate independently of each other.

a) Calculate the prob that an arriving airplane will be detected by atleast four radars.

b) Knowing that atleast three radars detected a given airplane what is the probability that the five radars detected this airplane

c) What is the smallest number n of radars that must be installed if we want an arriving airplane to be detected by atleast one radar with prob 0.9999?

a) X: the No. of radars that detect the airplane.

$$X \sim B(n, p) \quad n = 5, p = 0.9$$

$$P(X \geq 4) = p(4) + p(5) = 0.9185$$

b) $$P(X = 5 \mid X \geq 3) = \frac{P(X = 5 \cap X \geq 3)}{P(X \geq 3)} = \frac{P(X=5)}{P(X \geq 3)}$$

$$= 0.596.$$

(c) $X \sim B(n, p = 0.9)$ we seek smallest n such that $P(X \geq 1) = 0.9999$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{n}{0}(0.9)^0(0.1)^{n-0}$$

$$= 1 - (0.1)^n$$

$$\Rightarrow (0.1)^n = 1 - 0.9999$$

$$\Rightarrow (0.1)^n = 0.0001 = (0.1)^4$$

$$\Rightarrow \boxed{n_{min} = 4}$$

Ex. The joint p.d.f. of X & Y is given as

$$g(x,y) = \frac{e^{-(x+y)} x^3 y^4}{\Gamma 4 \, \Gamma 5}, \quad x > 0, y > 0$$

Find (i) p.d.f. of $\frac{X}{X+Y}$ (ii) $E(U)$ (iii) $V(U)$

Sol^n. Let $U = \frac{x}{x+y}$ and $v = x+y$

$$\Rightarrow x = uv, \quad y = v - x = v - uv = v(1-u)$$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v$$

$$0 < u < 1, \quad v > 0$$

$$p(u,v) = g(x,y)\,|J| = \frac{1}{\Gamma 4 \, \Gamma 5} e^{-v} (uv)^3 [v(1-u)]^4 \cdot v$$

$$= \frac{1}{\Gamma 4 \, \Gamma 5} e^{-v} v^8 u^3 (1-u)^4; \quad 0 \le u \le 1, \; v > 0$$

$$= \left(\frac{1}{\Gamma 9} e^{-v} v^8\right)\left(\frac{\Gamma 9}{\Gamma 4 \, \Gamma 5} u^3 (1-u)^4\right)$$

$$= p_1(v)\, p_2(u)$$

$$u = \frac{X}{X+Y} \sim \beta_1(4,5), \quad v \sim \Gamma 9$$

$$E(U) = \frac{4}{9}, \quad V(U) = \frac{2}{81}$$

variance $= \frac{\mu\nu}{(\mu+\nu)^2(\mu}$

$\beta_1(\mu,\nu)$

$f(x) = \frac{1}{\beta(\mu,\nu)} x^{\mu-1}(1-x$

Mean $= \frac{\mu}{\mu+\nu}$

variance $=$

$-(n-\frac{1}{2}) \mid n-\frac{1}{2}$

$$\cdots 2 = 2\sigma^2 \frac{(2n-1)}{2}$$

$$= \sigma^2(2n-1)$$

$$\Rightarrow \boxed{\mu_{2n} = \sigma^2(2n-1)\mu_{2n-2}} \text{ — recurrence relation}$$